

Intel Reasoning v2

End-to-End Plan & Contract — Spec only

Repo: prashanthkrishnan91/claude_financetracker_pk91 · Scope: v2/ · No code, no file edits.

1. Current Intel data-flow map

```
[provider layer]
v2/backend/app/services/market_data/* (yfinance fundamentals + history)
fetch_yfinance_fundamentals_sync() -> fundamentals dict
fetch_yfinance_history_sync()      -> price/return series
|
v
[feature layer]
v2/backend/app/services/intelligence/feature_engine.py -> FeatureSet
|
v
[deterministic intel]                                [LLM intel]
intelligence/thesis_mapper.py                        services/agents/per_ticker_analyst.py
map_to_thesis_inputs()                               runs Sonnet; emits analyst_verdict:
|                                                     { action, conviction, key_drivers,
|                                                     risks, primary_driver, risk_flag,
|                                                     action_reason, differentiation,
|                                                     generation_version: human_v2,
|                                                     analysis_source: live_llm/cache,
|                                                     used_fallback }
v                                                     |
intelligence/thesis_engine.py                        |
score_thesis() -> ScoreCard                          |
| (5 subscores, conviction band,                    |
| blended_data_quality, status)                     |
v                                                     |
intelligence/thesis_plain_english.py                 |
build_thesis_plain_english(card)                     |
-> { headline, *_label, caveats }                     |
|                                                     |
+-----+-----+-----+-----+-----+-----+-----+
|                                                     |
v                                                     |
[orchestration/persistence]                          |
recommendation_engine._persist                       |
writes recommendations rows (with agent_run_id)      |
writes agent_runs.allocation["_thesis_v2"] (per-ticker scorecards)
writes agent_insights row per ticker (analyst_verdict JSONB)
|
v
[card assembly]
recommendation_engine._compute_insight_cards
joins recs x analyst_verdict x _thesis_v2 (with fallback to latest_thesis_run)
normalize_reasoning_payload() -> canonical reasoning fields
_build_thesis_fields_for_card -> thesis_v2 + thesis_plain_english
emits InsightCard (models/recommendation.py)
|
v
[API] GET /api/v1/recommendations/ -> list[InsightCard]
|
v
[frontend]
v2/frontend/src/components/cards/AgentInsightCard.tsx
WHY      = primary_driver | analyst_drivers[0] | key_drivers[0]
RISK     = risk_flag      | analyst_risks[0]   | main_risks[0]
ACTION   = action_reason
ALT VIEW = differentiation (only when reasoning_schema_version in human_v2/compact_v1)
Footer chips: P&L, Ticker data quality, reasoning_source badge
Business read (thesis_plain_english) was previously rendered here -- now removed.
```

```
[Deploy v2]
services/deployment_engine.py + allocation_engine.py
  Owns allocation math. Today reads conviction/recommendation rows but does
  NOT consume thesis_v2 structurally. This is where reasoning_v2 deploy_signals
  will plug in later (additive only).
```

Two parallel reasoning streams exist (deterministic scorecard + LLM analyst verdict) and they are joined ad-hoc on the card. There is no single backend "reasoning object" that explains, cross-checks, or hands off to Deploy.

2. Why the Business read approach was the wrong product surface

bullet surfaced thesis_plain_english directly to amateur users while the underlying contract still produced INSUFFICIENT_DATA or near-identical labels for most live tickers (low ROIC/FCF coverage in yfinance, missing peer/own-history baselines, proxy trend_regime_score).

bullet the translator hardcoded prose patterns per status, so even after the directional-label fix every card collapsed into the same cautious paragraph.

bullet surfaced an unaudited fifth panel that competed with WHY/RISK/ACTION/ALT VIEW for attention without adding new signal — the LLM verdict already filled those four. Business read became visual noise that looked confident.

bullet Six PRs in the recent HANDOFF were point-fixes (run binding, tolerant ticker lookup, INSUFFICIENT_DATA branch, fallback run, copy cleanup, hide UI). None of them addressed the missing structural contract: there was no field-level proof that the deterministic scorecard agreed with, contradicted, or strengthened the LLM verdict before it reached the user.

bullet exposed backend metric vocabulary (conviction band, data quality, dimension labels) directly on the card — which violates the "amateur-investor friendly" copy invariant.

bullet Deploy was never wired to consume thesis_v2 anyway, so the user-visible Business read was a UI-first feature with no backend consumer pressure-testing the data shape.

Conclusion: the data needs to mature inside the backend (read by Deploy and used to strengthen WHY/RISK/ACTION/ALT VIEW) before re-earning a card surface.

3. Intel Reasoning v2 — design intent

A single deterministic-first reasoning object built per ticker by the backend that:

- 1 Fuses the deterministic scorecard with the LLM analyst verdict into one explainable structure.
- 2 Strengthens the existing WHY/RISK/ACTION/ALT VIEW fields the user already understands (no new card section).
- 3 Exposes a small, vetted set of deploy_signals — bands, postures, blockers — that Deploy can later read without learning Intel's metrics.
- 4 Is honest about missing data via a data_quality envelope; never fabricates.
- 5 Keeps backend-only metric names off the wire to the frontend in user-safe text fields.

Three behavioral guarantees:

bullet LLM may explain, summarize, or rephrase but must not invent metrics, scores, source claims, source facts, or allocations.

build numbers and bands are deterministic and traceable to evidence.

Deploy v2 keeps allocation math; Intel emits qualitative posture only.

4. reasoning_v2 contract — exact field schema

Top-level shape, attached to InsightCard (initially internal-only — see PR sequence). All fields are optional at the wire level so missing data degrades gracefully.

```
reasoning_v2: {
  "schema_version": "reasoning_v2.0",
  "ticker": "MSFT",

  "why":      { ...WhySection },
  "risk":     { ...RiskSection },
  "action":   { ...ActionSection },
  "alt_view": { ...AltViewSection },

  "confidence": { ...ConfidenceBlock },
  "deploy_signals": { ...DeploySignals },
  "evidence":    { ...EvidenceBlock },
  "data_quality": { ...DataQualityBlock }
}
```

4.1 why — strengthens WHY card field

```
{
  "user_text": "string <=180 chars, plain English, amateur-friendly, no metric jargon",
  "drivers":   ["string", ...] // up to 3, deterministic-first; LLM allowed to rephrase only
  "support":   "deterministic" | "analyst" | "both" | "none",
  "subscore_basis": ["quality" | "growth" | "valuation" | "momentum" | "risk", ...]
}
```

Build rule: if quality.published AND quality.score >= 70 -> first driver references business quality. If LLM primary_driver exists and is non-forbidden, prefer LLM phrasing but require deterministic backing. Otherwise emit deterministic template.

4.2 risk — strengthens RISK card field

```
{
  "user_text": "string <=180",
  "risks":     ["string", ...] // up to 3
  "severity":  "LOW" | "MEDIUM" | "HIGH",
  "support":   "deterministic" | "analyst" | "both" | "none",
  "subscore_basis": ["risk" | "valuation" | ...]
}
```

Severity rule: HIGH if risk.published AND risk.score < 40 OR analyst risk_flag matches material concentration/leverage patterns. MEDIUM if risk.score 40-60 or partial. LOW if risk.score >= 70 and analyst supports. UNKNOWN -> omit severity, force user_text to honest-uncertainty template.

4.3 action — strengthens ACTION card field

```
{
  "user_text": "string <=180", // what the user should DO this week
  "posture":   "ACCUMULATE" | "HOLD" | "TRIM" | "AVOID" | "WATCH",
  "horizon":   "short" | "medium" | "long" | null,
  "support":   "deterministic" | "analyst" | "both" | "none"
}
```

Action posture is NOT an allocation. It maps to recommendation action only when both deterministic conviction and analyst verdict agree; otherwise defaults to WATCH and is explicit about the disagreement in user_text.

4.4 alt_view — strengthens ALT VIEW card field

```
{
  "user_text": "string <=180",          // bear case / contrarian read
  "thesis":    "string <=220",
  "support":   "deterministic" | "analyst" | "both" | "none"
}
```

Sourced primarily from analyst differentiation + valuation/growth opposing dimension. If valuation looks expensive while quality is high, alt_view explicitly states the multiple risk.

4.5 confidence

```
{
  "conviction_band": "HIGH" | "MEDIUM" | "LOW" | "INSUFFICIENT_DATA",
  "conviction_score": 0-100 | null,
  "agreement":       "agree" | "disagree" | "partial" | "n/a",
  "agreement_note":  "string <=140"      // explains deterministic vs analyst alignment
}
```

agreement is a cross-check: deterministic conviction band vs analyst action mapped onto ACCUMULATE/HOLD/TRIM/AVOID. disagree -> posture is forced to WATCH and confidence.conviction_band is downgraded one step.

4.6 deploy_signals — Deploy v2's structured input (no allocation math)

```
{
  "conviction_band": "HIGH" | "MEDIUM" | "LOW" | "INSUFFICIENT_DATA",
  "risk_band":       "LOW" | "MEDIUM" | "HIGH" | "UNKNOWN",
  "data_quality_band": "HIGH" | "MEDIUM" | "LOW",
  "action_posture":   "ACCUMULATE" | "HOLD" | "TRIM" | "AVOID" | "WATCH",
  "watchlist_reason": "string | null", // null when not on watch
  "blockers":         ["insufficient_data" | "agreement_conflict" |
                       "stale_run" | "fallback_only" | "no_evidence", ...],
  "caveats":          ["string <=140", ...] // up to 3
}
```

Hard non-goals: no allocation amounts, no deploy percentages, no buy/sell sizes, no position targets. Deploy v2 reads bands/postures and runs its own allocation math.

Mapping rules (deterministic, table-driven):

bullet conviction_band = scorecard.conviction_band.

bullet risk_band = derived from risk subscore + analyst risk_flag presence; UNKNOWN if risk.published == False.

bullet data_quality_band: HIGH >= 0.75 blended_data_quality; MEDIUM >= 0.50; else LOW.

bullet action_posture = (conviction_band, agreement, risk_band) lookup table.

bullet blockers populated when any of: status == INSUFFICIENT_DATA, agreement == disagree, fallback run used (attached_via_latest_run), analyst_used_fallback == True, no provider data older than threshold.

4.7 evidence

```
{
  "deterministic": [
    {
      "key": "valuation",          // backend dimension name
      "score": 62.5,
      "data_quality": 0.71,
      "published": true,
      "inputs_used": ["forward_pe", "ps_ttm", "fcf_yield"],
      "inputs_missing": ["peer_ps_median", "ev_ebitda"]
    }, ...
  ]
}
```

```

    },
    "analyst": {
      "generation_version": "human_v2" | "compact_v1" | null,
      "analysis_source": "live_llm" | "cached_run" | "fallback" | null,
      "used_fallback": true | false,
      "citations": ["string", ...] // analyst-provided sources only
    },
    "provider": {
      "fundamentals_keys_present": ["pe", "forward_pe", "revenue_growth", ...],
      "fundamentals_keys_missing": ["roic_ttm", "fcf_margin_source", ...],
      "history_window_days": 252 | null,
      "as_of": "ISO-8601" | null
    }
  }
}

```

Every claim that surfaces in why/risk/action/alt_view/deploy_signals must be traceable to either evidence.deterministic (with published == True) or evidence.analyst.citations. The builder enforces this; unsupported claims are dropped.

4.8 data_quality

```

{
  "status": "READY" | "PARTIAL" | "INSUFFICIENT_DATA",
  "blended_data_quality": 0.0-1.0,
  "missing_inputs": ["string", ...],
  "stale": true | false,
  "as_of": "ISO-8601" | null,
  "user_text": "string <=140" // honest plain-English, no metric jargon
}

```

5. Source / evidence rules

- 1 Deterministic claims: only allowed when source subscore is published == True AND data_quality >= MIN_SUBSCORE_QUALITY (0.40). Inputs the score consumed must be enumerated in evidence.deterministic[*].inputs_used.
- 2 Analyst claims: allowed only when evidence.analyst.generation_version in {human_v2, compact_v1} AND used_fallback == False. Otherwise analyst text is dropped from user-safe fields and only its existence is recorded in evidence.
- 3 No invention: builder must refuse to emit user-safe text that references a dimension whose subscore is unpublished. The reasoning_contract_FORBIDDEN_PATTERNS regex must apply to all user_text strings before commit (no SMA/RSI/MACD/etc. leaking through).
- 4 Citations: evidence.analyst.citations is pass-through from analyst_verdict; never synthesized.
- 5 Deterministic over LLM when both are present and disagree on a single field.

6. Missing-data rules

- bullet status == INSUFFICIENT_DATA -> why.user_text, risk.user_text, action.user_text all use honest-uncertainty templates and action.posture = "WATCH". No bands beyond INSUFFICIENT_DATA / UNKNOWN.
- bullet status == PARTIAL -> unpublished dimensions are simply omitted from subscore_basis; user_text can still be emitted from published dimensions.
- bullet Provider as-of older than freshness window (configurable, default 7 days for fundamentals, 2 trading days for prices) -> data_quality.stale = true and deploy_signals.blockers includes "stale_run".

bullet Analyst-only claims with no deterministic backing -> allowed in alt_view only (user-flagged as "alternative read"), never in action.

bullet Empty list / null is preferred over guessed-but-honest copy; the frontend already handles missing gracefully.

7. Deploy-signal rules (forward contract)

Deploy v2 may consume only reasoning_v2.deploy_signals. Explicit:

bullet Allowed: conviction_band, risk_band, data_quality_band, action_posture, watchlist_reason, blockers, caveats.

bullet NOT allowed: any allocation amount, deploy percent, target weight, dollar size, fractional share count, time-in-market, lot suggestion.

bullet Deploy must treat reasoning_v2 as advisory metadata; allocation math stays in allocation_engine.py and deployment_engine.py.

bullet Decision logs remain deterministic and continue to read from existing fields, not from reasoning_v2 (so no LLM dependency leaks into decision history).

bullet Re-using reasoning_v2 across runs requires deploy to check data_quality.as_of and data_quality.stale — Deploy must skip stale signals.

8. Test plan

PR 1 (builder) tests live in v2/backend/tests/test_reasoning_v2_builder.py:

- 1 Schema shape — top-level keys exact, all nested keys present (with optional values), schema_version stamped.
- 2 Deterministic-only ticker — analyst_verdict missing -> builder emits support="deterministic" everywhere, action.posture obeys lookup table, no analyst citations, evidence.analyst all-null.
- 3 Analyst-only ticker — scorecard INSUFFICIENT_DATA + valid human_v2 verdict -> user_text safe templates, posture forced to WATCH, blockers includes insufficient_data, alt_view may carry analyst differentiation.
- 4 Both present, agree — deterministic HIGH conviction + analyst BUY -> confidence.agreement="agree", deploy_signals.action_posture="ACCUMULATE".
- 5 Both present, disagree — deterministic LOW + analyst BUY -> posture forced to WATCH, agreement="disagree", conviction_band downgraded one step, blockers includes agreement_conflict.
- 6 Forbidden language scrub — analyst verdict containing "SMA20"/"RSI"/"momentum" must be sanitized out of every user_text; raw fields preserved only inside evidence.
- 7 Stale provider as_of -> data_quality.stale=true, blockers includes stale_run.
- 8 Fallback-only analyst (used_fallback=true) -> analyst text excluded from why/risk/action; recorded in evidence.analyst.used_fallback.
- 9 PARTIAL status with mixed published dimensions -> why/risk emitted only for published dimensions; subscore_basis enumerates only published ones.
- 10 Determinism — same inputs -> byte-equal JSON output (excluding as_of if it's pass-through).
- 11 No allocation leakage — assert no key path produces a numeric allocation, percent target, share size, dollar amount.
- 12 Data quality math — blended_data_quality matches ScoreCard.blended_data_quality exactly; data_quality_band thresholds verified at 0.50 / 0.75 boundaries.

- 13 Evidence traceability — every subscore_basis entry exists in evidence.deterministic with published=True.
- 14 Snapshot tests for 3 representative ticker shapes (READY/PARTIAL/INSUFFICIENT_DATA).

PR 2 wiring tests added to test_recommendation_engine.py:

bullet Card response includes reasoning_v2 when _thesis_v2 present; omitted/null otherwise.

bullet WHY/RISK/ACTION/ALT VIEW source preference order is preserved (no regression in existing AgentInsightCard tests).

9. PR sequence

PR	Scope	Files	Usage	Risk
PR 1	Backend reasoning_v2 builder + tests; wire into agent_runs.allocation["_reasoning_v2"] only (NOT on InsightCard yet)	new: intelligence/reasoning_v2_builder.py, tests/test_reasoning_v2_builder.py; small edit: recommendation_engine.py	Medium	Low — additive, dormant
PR 2	Add reasoning_v2: Optional[dict] to InsightCard; project from _reasoning_v2 map; internally fortify WHY/RISK/ACTION/ALT VIEW selection; no new UI section	models/recommendation.py, recommendation_engine.py, tests	Medium	Low–Medium
PR 3 (optional)	Frontend display refinement only if PR 2 telemetry shows clearly improved WHY/RISK/ACTION/ALT VIEW quality across 5+ live tickers; possibly an "agreement" badge	AgentInsightCard.tsx, test	Low	Low — UI cap = 2 files
PR 4	Deploy v2 reads reasoning_v2.deploy_signals; allocation math unchanged; gated by feature flag with side-by-side comparison	deployment_engine.py, allocation_engine.py, tests	Medium	Medium

Strict invariant: PR 3 does NOT re-enable Business read or thesis_plain_english rendering. The dormant thesis_plain_english stays dormant.

10. First implementation prompt for Claude Sonnet (PR 1)

Outside the prompt block:

bullet Model: Claude Sonnet

bullet Chat: new

bullet Usage estimate: Medium · extra cost risk: Low · Why: one new module, one new test file, one small wire-up in recommendation engine; no UI, no SQL.

Task:

Implement Intel Reasoning v2 PR 1 -- backend-only deterministic builder

that fuses the existing thesis scorecard and analyst verdict into a single reasoning_v2 object per ticker. Persist it on agent_runs.allocation under "_reasoning_v2" alongside the existing "_thesis_v2". Do NOT add it to the InsightCard response. Do NOT touch the frontend. Do NOT re-enable thesis_plain_english rendering.

Primary edit targets:

- v2/backend/app/services/intelligence/reasoning_v2_builder.py (new)
 - v2/backend/tests/test_reasoning_v2_builder.py (new)
 - v2/backend/app/services/recommendation_engine.py
- Only the orchestration site that already writes "_thesis_v2" into agent_runs.allocation. Add a sibling write for "_reasoning_v2".

Fallback reads only if blocked:

- v2/backend/app/services/intelligence/score_schema.py
- v2/backend/app/services/intelligence/thesis_engine.py
- v2/backend/app/services/intelligence/thesis_mapper.py
- v2/backend/app/services/agents/per_ticker_analyst.py
- v2/backend/app/services/reasoning_contract.py

Contract (must implement exactly this):

```
def build_reasoning_v2(  
    *,  
    ticker: str,  
    scorecard: ScoreCard | dict | None,  
    analyst_verdict: dict | None,  
    provider_meta: dict | None = None,  
) -> dict
```

Output schema (top-level keys, all required, values may be None):

schema_version="reasoning_v2.0", ticker, why, risk, action, alt_view, confidence, deploy_signals, evidence, data_quality.

Section rules (summarised -- see plan):

- why/risk/action/alt_view: { user_text, support, ... }
- user_text <=180 chars, plain English, must NOT contain any pattern from reasoning_contract._FORBIDDEN_PATTERNS. Strip if it does.
- action.posture in {ACCUMULATE,HOLD,TRIM,AVOID,WATCH}; map from (conviction_band, agreement, risk_band) lookup table; force WATCH on disagree or INSUFFICIENT_DATA.
- alt_view may use analyst differentiation; everything else must trace to a published deterministic subscore or analyst citation.
- deploy_signals: NEVER emit allocation amounts/percents/sizes.
- evidence.deterministic enumerates each subscore with score, data_quality, published, inputs_used, inputs_missing.
- evidence.analyst captures generation_version, analysis_source, used_fallback, citations (pass-through only).
- data_quality.status / blended_data_quality / missing_inputs / stale / as_of / user_text.

Determinism:

- Pure function. No IO. No LLM. No DB. No yfinance. No randomness.
- Same inputs -> same JSON output.

Tests (must include all of these):

1. schema shape
2. deterministic-only ticker
3. analyst-only ticker
4. both-present-agree
5. both-present-disagree (forces WATCH, downgrades band, adds blocker)
6. forbidden language scrub
7. stale provider as_of -> blockers contains "stale_run"
8. fallback-only analyst (used_fallback=True) excludes analyst text
9. PARTIAL status emits only published dimensions

10. determinism (same input -> same output)
11. no allocation leakage (regex assert)
12. blended_data_quality math at 0.50 / 0.75 boundaries
13. evidence traceability (subscore_basis subset of evidence.deterministic published)
14. snapshot tests for READY / PARTIAL / INSUFFICIENT_DATA tickers

Wire-up in recommendation_engine.py:

- At the existing site that builds "_thesis_v2" per-ticker for agent_runs.allocation, also build the matching reasoning_v2 dict and store it under allocation["_reasoning_v2"][ticker].
- Wrap in try/except: failure must NOT break the run. Log "reasoning_v2.build_failed".
- Do NOT add reasoning_v2 to the InsightCard pydantic model.
- Do NOT change any user-visible field.

Discovery budget:

- no find/grep/glob for initial exploration
- read primary + tests first
- fallback files only if blocked

Timeout budget:

- checkpoint before timing out
- max two continues, then fresh chat / checkpoint compression

Constraints:

- no refactors of thesis_engine, thesis_mapper, thesis_plain_english, reasoning_contract, or per_ticker_analyst
- no Supabase SQL, no migration
- no LLM prompt change
- no frontend change
- no Deploy change
- minimal patch

MANDATORY:

- Update docs/ai/HANDOFF.md in the same change describing PR 1 scope, the new module, and that reasoning_v2 is NOT yet exposed on InsightCard.
- After opening the PR, stop. Do not propose the next implementation prompt.

Output:

- files
- tests
- handoff update summary

11. Risks and open questions

Risks

- bullet Provider data sparsity (yfinance) means many tickers will still emit INSUFFICIENT_DATA. PR 1 will mostly emit honest-uncertainty objects; we accept that and use it as the proof gate for PR 3.
- bullet Cross-check (agreement) could downgrade currently-shipping cards from BUY to WATCH if PR 2 wires it strictly. PR 2 must keep recommendations.action untouched and only adjust internal selection of WHY/RISK/ACTION text. Action mapping change is a Deploy concern (PR 4).
- bullet reasoning_v2 adds JSON volume to agent_runs.allocation. Estimate ~1-2 KB per ticker; for ~30 tickers that's ~50 KB additional JSONB. Within Supabase row limits but worth confirming on first run.
- bullet Forbidden-language scrub could over-strip valid plain-English text ("trend" appears in _FORBIDDEN_PATTERNS). Builder must use deterministic templates that already avoid these patterns, not rely on post-hoc stripping.

Open questions

- 1 Should data_quality.as_of come from provider snapshot timestamp or from agent_runs.finished_at?
Recommend: provider-level for fundamentals (as_of_fundamentals) and run-level for the overall reasoning object. Confirm during PR 1.
- 2 Should agreement="disagree" block the recommendation persistence in _persist, or only adjust reasoning_v2?
Recommend: NO blocking in PR 1-2. Surface only via deploy_signals.blockers and let PR 4 decide.
- 3 Should deploy_signals.action_posture map 1:1 to recommendations.action, or stay independent?
Recommend: independent. Recommendation action stays driven by today's logic; posture is advisory.
- 4 PR 2 — should reasoning_v2 ship on InsightCard wire even if dormant in UI? See section 12.
- 5 Future: do we want a per-run aggregate reasoning_v2 (portfolio-level conviction/risk bands)? Defer to a later PR; out of scope here.

12. Recommendation: PR 1 backend-only or response-visible?

PR 1 must be backend-only (persist to agent_runs.allocation["_reasoning_v2"], not on InsightCard).
Reasons:

- bulletThe product lesson from Business read was that exposing reasoning data prematurely created low-value UI before the contract was proven on real runs. Mirroring that mistake on reasoning_v2 re-runs the same risk.
- bulletagent_runs.allocation already carries _thesis_v2 and is the natural sibling location. The frontend cannot accidentally render it.
- bulletlets us inspect 1-2 weeks of live runs (logs + DB queries) to verify agreement, blockers, data_quality.stale behave as designed before letting Deploy or the card touch them.
- bulletkeeps PR 1 truly minimal-scope (one new module, one new test file, one small wire-up) and avoids contract churn on InsightCard until PR 2 lands intentionally.
- bulletPR 2 then becomes a controlled, additive expansion: project _reasoning_v2 onto InsightCard.reasoning_v2 and use it internally to refine WHY/RISK/ACTION/ALT VIEW selection — still no new UI section.

Final flags

Question	Answer
Supabase SQL expected	No. _reasoning_v2 lives inside the existing agent_runs.allocation JSONB column. No schema, no migration, no RLS change.
LLM behavior expected to change	No. No prompt, model, or analyst-pipeline change. The builder consumes whatever analyst_verdict already exists.
Frontend/UI expected to change	No in PR 1. PR 2 adds an optional reasoning_v2 field to the API response (still no new UI section). PR 3 (optional, gated) is the only frontend-touching PR.
Deploy expected to change	No in PR 1-3. Deploy is touched only in PR 4, after live evidence proves deploy_signals are useful.
HANDOFF.md / README.md	Not edited by this planning artifact. HANDOFF.md will be updated inside PR 1.

PR-summary block (for PR 1 author)

Root cause/plan: Replace one-symptom-at-a-time Intel patching with a structured backend reasoning_v2 contract that fuses the deterministic scorecard with the analyst verdict, strengthens WHY/RISK/ACTION/ALT VIEW, and exposes deploy-safe bands/postures (no allocations) for Deploy v2 to consume later.

Files changed: see PR.

Tests: 14 unit tests for the reasoning_v2 builder (PR 1).

Risks: Sparse yfinance coverage will produce many INSUFFICIENT_DATA reasoning_v2 objects; agreement downgrades must not change recommendations.action in PR 1-2; ~1-2 KB JSONB growth per ticker.

Supabase SQL: No

HANDOFF.md edited: Yes -- PR 1 scope, new builder module, dormant on InsightCard.

README.md edited: No -- no public/setup behavior change.